THE FIRST COHOMOLOGY OF THE UNITS OF A LOCALLY RINGED SITE IS THE PICARD GROUP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed. The author writes 'Then may use Lemma' and 'a an ideal' elsewhere in the batch; both slips are kept verbatim where they occur.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; no macro was invented or redefined.

Editorial note. Similarity note: this chapter also states lemma-torsors-h1 (03AJ), which classifies O^*-torsors. The present card counts the Picard-group identification only and uses 03AJ strictly as a navigation pointer.

Allowed standard prerequisites (editorial).

- Invertible O-modules and their local triviality of rank one. Modules on Sites, Lemma labelled sites-modules-lemma-invertible-is-locally-free-rank-1 (canonical Stacks tag 0B8Q); invoked at proof lines 543-544. Supplies that the sheaf of nondegenerate sections of an invertible module has sections locally; the torsor construction on top of it is retained in the counted span.
- The bijection between O^*-torsors and H^1 of the units. Same chapter, Lemma labelled lemma-torsors-h1 (source line 291, Stacks tag 03AJ, already counted separately); invoked at proof lines 560-563. Supplies the identification of the set of torsors with H^1(C, O^*); the map from invertible sheaves to torsors is retained in the counted span.
- Trivial torsors have a global section. Same chapter, Lemma labelled lemma-trivial-torsor (source line 278, Stacks tag 03AI); invoked at proof lines 567-568. Supplies the injectivity step, i.e. that a trivial associated torsor means the sheaf is free; the surjectivity construction is retained in the counted span.

Source reference. Original lemma 0B8Q: sources/stacks-chow/tags.txt, source lines 14148–14148. Cross-chapter standard prerequisite invoked once at proof lines 543-544 for local existence of nondegenerate sections; the torsor construction built on it is retained in the counted span.

Source reference. Original lemma 03AJ: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 291–291. Same-chapter pointer invoked at proof lines 560-563 for the identification of torsors with H^1(C, O^*); the map from invertible sheaves to torsors and its bijectivity are retained in the counted span.

Source reference. Original lemma 03AI: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 278–278. Same-chapter pointer invoked at proof

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

lines 567-568 for the injectivity step; the construction used for surjectivity is retained in the counted span.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a locally ringed site. There is a canonical isomorphism*

$$H^1(\mathcal{C}, \mathcal{O}^*) = \mathrm{Pic}(\mathcal{O}).$$

of abelian groups.

Proof. Let $\mathcal{L}$ be an invertible $\mathcal{O}$-module. Consider the presheaf $\mathcal{L}^*$ defined by the rule

$$U \longmapsto \{s \in \mathcal{L}(U) \text{ such that } \mathcal{O}_U \xrightarrow{s\cdot -} \mathcal{L}_U \text{ is an isomorphism}\}$$

This presheaf satisfies the sheaf condition. Moreover, if $f \in \mathcal{O}^*(U)$ and $s \in \mathcal{L}^*(U)$, then clearly $fs \in \mathcal{L}^*(U)$. By the same token, if $s, s' \in \mathcal{L}^*(U)$ then there exists a unique $f \in \mathcal{O}^*(U)$ such that $fs = s'$. Moreover, the sheaf $\mathcal{L}^*$ has sections locally by Modules on Sites, Lemma 0B8Q. In other words we see that $\mathcal{L}^*$ is a $\mathcal{O}^*$-torsor. Thus we get a map

$$\begin{array}{c}\text{set of invertible sheaves on } (\mathcal{C}, \mathcal{O}) \\ \text{up to isomorphism}\end{array} \longrightarrow \begin{array}{c}\text{set of } \mathcal{O}^*\text{-torsors} \\ \text{up to isomorphism}\end{array}$$

We omit the verification that this is a homomorphism of abelian groups. By Lemma 03AJ the right hand side is canonically bijective to $H^1(\mathcal{C}, \mathcal{O}^*)$. Thus we have to show this map is injective and surjective.

Injective. If the torsor $\mathcal{L}^*$ is trivial, this means by Lemma 03AI that $\mathcal{L}^*$ has a global section. Hence this means exactly that $\mathcal{L} \cong \mathcal{O}$ is the neutral element in $\mathrm{Pic}(\mathcal{O})$.

Surjective. Let $\mathcal{F}$ be an $\mathcal{O}^*$-torsor. Consider the presheaf of sets

$$\mathcal{L}_1 : U \longmapsto (\mathcal{F}(U) \times \mathcal{O}(U))/\mathcal{O}^*(U)$$

where the action of $f \in \mathcal{O}^*(U)$ on (s, g) is $(fs, f^{-1}g)$. Then $\mathcal{L}_1$ is a presheaf of $\mathcal{O}$-modules by setting $(s, g)+(s', g') = (s, g+(s'/s)g')$ where s'/s is the local section f of $\mathcal{O}^*$ such that $fs = s'$, and $h(s, g) = (s, hg)$ for h a local section of $\mathcal{O}$. We omit the verification that the sheafification $\mathcal{L} = \mathcal{L}_1^\#$ is an invertible $\mathcal{O}$-module whose associated $\mathcal{O}^*$-torsor $\mathcal{L}^*$ is isomorphic to $\mathcal{F}$. □